



Nagar Yuvak Shikshan Sanstha's
YESHWANTRAO CHAVAN COLLEGE OF ENGINEERING, NAGPUR
(An Autonomous Institute Affiliated to Rashtrasant Tukadoji Maharaj Nagpur University) Hingna
Road, Wanadongri, Nagpur - 441110
Accredited with A++ Grade by NAAC

DEPARTMENT OF COMPUTER TECHNOLOGY

Vision

To be a well-known center for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary arena by developing problem-solving skills through emerging technologies.

Project Title: Banking Management System

This project report is submitted to

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution Affiliated to Rashtra Sant Tukdoji Maharaj Nagpur University)

DBMS Lab

by

Shlok Bhoyar

Sharwari Kapse

Shamik Modak

Samyak Pusate

Under the guidance of

Sharayu Kewale



Nagar Yuvak Shikshan Sanstha's
YESHWANTRAO CHAVAN COLLEGE OF ENGINEERING, NAGPUR
(An Autonomous Institute Affiliated to Rashtrasant Tukadoji Maharaj Nagpur University) Hingna
Road, Wanadongri, Nagpur - 441110
Accredited with A++ Grade by NAAC

DEPARTMENT OF COMPUTER TECHNOLOGY

Vision

To be a well-known center for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary arena by developing problem-solving skills through emerging technologies.

Course Code:(23ADS1406)	Course Name: DBMS Lab
Name of Students: Shlok Bhoyar Sharwari Kapse Shamik Modak Samyak Pusate	Semester/Section: Even: 2026-27 5-CT-A
Roll No.: 70 24 67 66	Enroll No.: 24071005 24071004 24070636 24070572

Introduction

The Banking Management System is a web-based application designed to manage basic banking activities digitally.

The system allows users to create and manage bank accounts, view account details, perform transactions, and check transaction history. It provides a simple and user-friendly interface for managing banking information.

The main purpose of this project is to understand how a Database Management System (DBMS) can be used to store, manage, and retrieve banking data efficiently



DEPARTMENT OF COMPUTER TECHNOLOGY

Vision

To be a well-known center for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

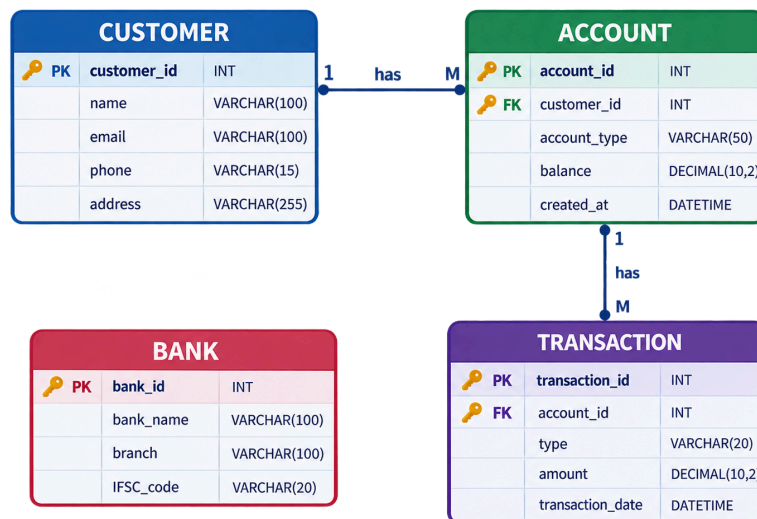
Mission

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary arena by developing problem-solving skills through emerging technologies.

Tools & Technologies

- **Frontend:** HTML, CSS, Bootstrap
- **Backend:** Node.js, Express.js
- **Database:** MySQL
- **Template Engine:** EJS
- **Programming Language:** JavaScript
- **Development Tool:** Visual Studio Code
- **Version Control:** Git and GitHub

E-R Model





Nagar Yuvak Shikshan Sanstha's
YESHWANTRAO CHAVAN COLLEGE OF ENGINEERING, NAGPUR
(An Autonomous Institute Affiliated to Rashtrasant Tukadoji Maharaj Nagpur University) Hingna
Road, Wanadongri, Nagpur - 441110
Accredited with A++ Grade by NAAC

DEPARTMENT OF COMPUTER TECHNOLOGY

Vision

To be a well-known center for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary arena by developing problem-solving skills through emerging technologies.

Database Schema

Table	Column	Data Type	Key / Constraint
Customer	customer_id	INT	PK, AUTO_INCREMENT
	name	VARCHAR(100)	NOT NULL
	email	VARCHAR(100)	UNIQUE
	phone	VARCHAR(15)	—
	address	VARCHAR(255)	—
Account	account_id	INT	PK, AUTO_INCREMENT
	customer_id	INT	FK → Customer
	account_type	VARCHAR(30)	NOT NULL
	balance	DECIMAL(12,2)	DEFAULT 0
	created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP
Transaction	transaction_id	INT	PK, AUTO_INCREMENT
	account_id	INT	FK → Account
	type	VARCHAR(20)	NOT NULL
	amount	DECIMAL(12,2)	NOT NULL
	transaction_date	DATETIME	DEFAULT CURRENT_TIMESTAMP
Bank	bank_id	INT	PK, AUTO_INCREMENT
	bank_name	VARCHAR(100)	NOT NULL
	branch	VARCHAR(100)	—
	IFSC_code	VARCHAR(20)	UNIQUE



Nagar Yuvak Shikshan Sanstha's
YESHWANTRAO CHAVAN COLLEGE OF ENGINEERING, NAGPUR
(An Autonomous Institute Affiliated to Rashtrasant Tukadoji Maharaj Nagpur University) Hingna
Road, Wanadongri, Nagpur - 441110
Accredited with A++ Grade by NAAC

DEPARTMENT OF COMPUTER TECHNOLOGY

Vision

To be a well-known center for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary arena by developing problem-solving skills through emerging technologies.

Physical View of the Database

The physical view describes how the database is actually stored and accessed by the MySQL DBMS. In this project, the database is implemented in MySQL and contains relational tables. Each row represents one record and each column represents an attribute. Primary keys are indexed for unique identification, while foreign keys maintain relationships between tables.

DBMS: MySQL

Storage structure: Relational tables with rows and columns

Primary keys: Indexed unique identifiers for each table

Foreign keys: Used to connect Customer → Account and Account → Transaction

Numeric money values: DECIMAL(12,2) to avoid unnecessary floating-point rounding

Dates: DATETIME fields store creation and transaction time

Access: The Node.js/Express.js backend sends SQL queries to MySQL



Nagar Yuvak Shikshan Sanstha's
YESHWANTRAO CHAVAN COLLEGE OF ENGINEERING, NAGPUR
(An Autonomous Institute Affiliated to Rashtrasant Tukadoji Maharaj Nagpur University) Hingna
Road, Wanadongri, Nagpur - 441110
Accredited with A++ Grade by NAAC

DEPARTMENT OF COMPUTER TECHNOLOGY

Vision

To be a well-known center for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary arena by developing problem-solving skills through emerging technologies.

Screenshots of Implementation

The top screenshot displays the 'Add Customer' form in the BankEase application. The form includes fields for Name, Email, Phone (10 digits), Address, and Date of Birth (mm/dd/yyyy). The background shows a table of existing customers with columns for ID, Name, Phone, Address, Date of Birth, and Actions (Edit, Delete).

The bottom screenshot displays the 'Create Account' form. The form includes a dropdown for Customer selection, a dropdown for Account Type (Savings, Current), a field for Opening Balance (INR), and a field for Opening Date (mm/dd/yyyy). The background shows a table of existing accounts with columns for ACCOUNT ID, CUSTOMER, ACCOUNT TYPE, OPENING BALANCE (INR), STATUS, and ACTIONS (View, Edit).



Nagar Yuvak Shikshan Sanstha's
YESHWANTRAO CHAVAN COLLEGE OF ENGINEERING, NAGPUR
(An Autonomous Institute Affiliated to Rashtrasant Tukadoji Maharaj Nagpur University) Hingna
Road, Wanadongri, Nagpur - 441110
Accredited with A++ Grade by NAAC

DEPARTMENT OF COMPUTER TECHNOLOGY

Vision

To be a well-known center for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary arena by developing problem-solving skills through emerging technologies.

The screenshot shows the 'Loans' management page in the BankEase application. It features a sidebar with navigation options like Dashboard, Customers, Accounts, Transactions, Loans, Loan Payments, Branches, and Employees. The main content area has a search bar and a table of loan records. Each record includes fields for Loan ID, Customer, Type, Amount, Interest %, Tenure, Status, Applied date, and Update Status, with corresponding 'Save' buttons.

LOAN ID	CUSTOMER	TYPE	AMOUNT (INR)	INTEREST %	TENURE	STATUS	APPLIED	UPDATE STATUS
6	Sanjay Verma	Vehicle	₹ 5,20,000	9.10	48 months	Pending	2024-03-10	Pending Save
5	Arjun Mehta	Home	₹ 18,00,000	8.75	180 months	Rejected	2024-01-28	Rejected Save
4	Ananya Iyer	Personal	₹ 1,50,000	12.00	24 months	Approved	2024-02-15	Approved Save
3	Sneha Reddy	Education	₹ 4,00,000	7.90	48 months	Pending	2024-03-01	Pending Save
2	Priya Patel	Vehicle	₹ 6,50,000	9.25	60 months	Approved	2024-02-02	Approved Save
1	Rahul Sharma	Home	₹ 25,00,000	8.50	240 months	Approved	2024-01-20	Approved Save

The screenshot shows the 'Employees' management page in the BankEase application. It features a sidebar with navigation options like Dashboard, Customers, Accounts, Transactions, Loans, Loan Payments, Branches, and Employees. The main content area has a search bar and a table of employee records. Each record includes fields for ID, Name, Role, Branch, Email, and Actions (Edit/Delete).

ID	NAME	ROLE	BRANCH	EMAIL	ACTIONS
10	Suresh Naidu	Cashier	Banjara Hills Branch (Hyderabad)	suresh.naidu@bank.example	Edit Delete
9	Fatima Begum	Loan Officer	Banjara Hills Branch (Hyderabad)	fatima.begum@bank.example	Edit Delete
8	Ankit Chatterjee	Clerk	Salt Lake Branch (Kolkata)	ankit.chatterjee@bank.example	Edit Delete
7	Karthik Subramanian	Branch Manager	T Nagar Branch (Chennai)	karthik.s@bank.example	Edit Delete
6	Divya Shetty	Cashier	Koramangala Branch (Bengaluru)	divya.shetty@bank.example	Edit Delete
5	Harish Rao	Loan Officer	Koramangala Branch (Bengaluru)	harish.rao@bank.example	Edit Delete
4	Rakesh Yadav	Clerk	Connaught Place Branch (Delhi)	rakesh.yadav@bank.example	Edit Delete
3	Pooja Malhotra	Branch Manager	Connaught Place Branch (Delhi)	pooja.malhotra@bank.example	Edit Delete
2	Farhan Shaikh	Cashier	Andheri West Branch (Mumbai)	farhan.shaikh@bank.example	Edit Delete
1	Neha Kulkarni	Branch Manager	Andheri West Branch (Mumbai)	neha.kulkarni@bank.example	Edit Delete



Nagar Yuvak Shikshan Sanstha's
YESHWANTRAO CHAVAN COLLEGE OF ENGINEERING, NAGPUR
(An Autonomous Institute Affiliated to Rashtrasant Tukadoji Maharaj Nagpur University) Hingna
Road, Wanadongri, Nagpur - 441110
Accredited with A++ Grade by NAAC

DEPARTMENT OF COMPUTER TECHNOLOGY

Vision

To be a well-known center for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary arena by developing problem-solving skills through emerging technologies.

Results

- Users can manage bank account details.
- Banking transactions can be recorded and viewed.
- Transaction history can be displayed.
- Data can be stored and retrieved from the MySQL database.
- The system provides a simple and user-friendly interface.

Conclusion

The Banking Management System successfully demonstrates how a DBMS can be used to manage banking data. The project provides basic features such as account management, transactions, and transaction history.

Through this project, we learned how to connect a web application with a database and perform operations such as **INSERT, SELECT, UPDATE, and DELETE**. It also helped us understand the practical use of database management in a real-world application.